Learning route

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The SSD bundled with the product comes with a preinstalled system and case environments. **Before the first use, please check the bootloader version of the mainboard: factory mainboards boot with JetPack 6.2 by default, while the SSD currently bundled with the product runs JetPack 7.2.1, so you must first update the QSPI/UEFI bootloader as described in Chapter 2, and then install the SSD for booting.**

After the bootloader has been updated and the system has booted successfully from the bundled SSD, you generally do not need to flash the system again or to rebuild the preinstalled environments by repeating the tutorials; just run the commands required by each case. Unless there is an explicit requirement, beginners should not directly perform the clean system flash, system upgrade or reinstallation steps, so as not to overwrite the preinstalled environments.

Chapter introduction

It is recommended to study in the following order; the chapter numbers follow the current tutorial directory.

Chapter 1 Quick start

Introduces the preparation for first use, system information, network connection and remote access. Read the bootloader update instructions in "Quick start" before the first boot.

Chapter 2 Mainboard basics

Introduces mainboard hardware, bootloader updates, system backup and restore, system flashing and SSD expansion. Beginners usually only need to complete the applicable bootloader update and expand the SSD after entering the preinstalled system; flashing a clean system, restoring images and flashing your own SSD are on-demand operations.

Note:

1. Flashing the Jetson clean system and the component environment are optional operations, not part of the first-boot flow for the bundled SSD. The current JetPack 7.2.1 bootloader update and reinstalling a

clean system are two different operations; tell them apart by following the corresponding tutorial. Reinstallation may overwrite the data and the preinstalled case environments on the SSD.

2. The factory image may have been compressed. After booting successfully from the bundled SSD, you can check and use the remaining SSD space according to the SSD expansion tutorial in Chapter 2.

Chapter 3 Basic usage

Mainly introduces the basic Ubuntu operations and common tools of the current system; the exact distribution depends on the corresponding system version and the tutorial description.

Chapter 4 GPIO control

Mainly introduces GPIO pin control, serial port and I2C communication on the Jetson Orin series mainboards.

Chapter 5 Vision basics

Mainly introduces the simple invocation of CSI cameras and USB cameras; users can verify whether the camera works normally with the cases in this tutorial.

Chapter 6 OpenCV basics

Mainly introduces common OpenCV image processing and simple examples that provide camera invocation.

Chapter 7 Advanced vision

Introduces the vision development environment and advanced cases such as DeepStream, TAO, YOLO, MediaPipe and QR code recognition. Different sub-tutorials may correspond to different JetPack, CUDA or framework versions; check the applicable environment before you start, and do not install the dependencies of an older version directly onto the current system.

Chapter 8 Docker basics

Mainly introduces the basic Docker commands, methods and script startup methods. The ROS1 Melodic environment runs in a Docker container, and the related cases enter the container as required by the tutorial.

Chapter 9 ROS2-Jazzy

Mainly introduces the basics of ROS 2 Jazzy, tools and robot vision cases. The ROS2 environment runs on the host, and camera and other hardware support depends on the description of the specific case.

Chapter 10 Large model basic course

Introduces local large models, Ollama, model deployment and application basics.

Chapter 11 Large model application practice

Introduces the configuration and use of online and offline large models in text, speech and multimodal applications. Online services require a network connection and valid credentials; offline inference speed is affected by mainboard performance.

Chapter 12 NVIDIA Isaac ROS

Introduces the features and cases of NVIDIA Isaac ROS. This chapter has specific JetPack, ROS 2, Docker image and hardware requirements; check the environment baseline listed in the tutorial before you start.

Chapter 13 OpenClaw deployment and basic usage

Introduces the deployment and basic usage of OpenClaw.

Chapter 14 OpenClaw extension package advanced development

Introduces the advanced development of the OpenClaw extension package.